Supplemental Material:
Movies for "The Elastic Perimeter Disk Model"

Research Note

(Dated: April 22, 2026)

This supplemental material provides six movies of two-disk squeeze simulations at fixed $b = 0.2$, $N = 32$, $\varepsilon_{\mathrm{ref}} = 0.08$, illustrating the squishiness axis as q is varied from area-compressible to nearly incompressible. In all movies: the two EPD disks (blue/orange perimeter nodes) are squeezed between rigid horizontal walls (gray); the color of each node encodes the local contact force magnitude; the title shows current per-particle strain ε_p and instantaneous ν_{meas}.

All movies are provided as animated GIFs in the `movies/` subdirectory.

MOVIE S1 — FULL SQUISHINESS SWEEP (COMPARISON)

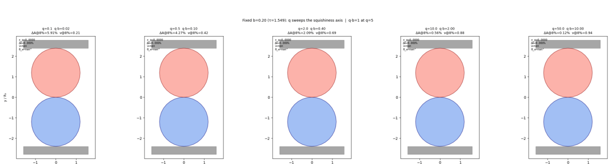


FIG. 1. **Movie S1** (`movies/S1_comparison_q_sweep.gif`): Side-by-side comparison of five q values ($q \in \{0.1, 0.5, 2, 10, 50\}$, left to right) at fixed $b = 0.2$, all compressed to the same wall displacement. The visual transition from strongly flattened area-compressible shapes (left) to laterally bulging nearly-incompressible shapes (right) illustrates the full squishiness axis. Still shown: initial configuration ($\varepsilon_p = 0$).

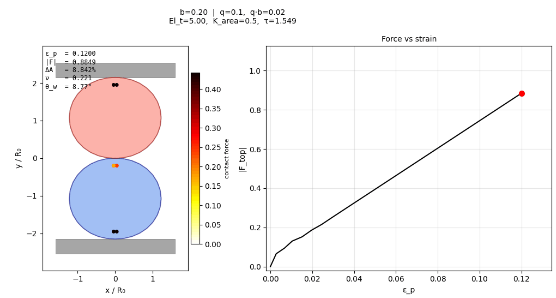


FIG. 2. **Movie S2** (`movies/S2_q0p1_area_compressible.gif`): Single squeeze simulation at $q = 0.1$ ($q \cdot b = 0.02$, $\nu_{\mathrm{meas}} \approx 0.21$, $\Delta A \approx 5.9\%$). The area penalty is weak; the disk compresses largely in place with little lateral expansion. The flattened top and bottom contact zones are clearly visible; the sides remain nearly straight. Still shown: maximum compression ($\varepsilon_p \approx 0.14$).

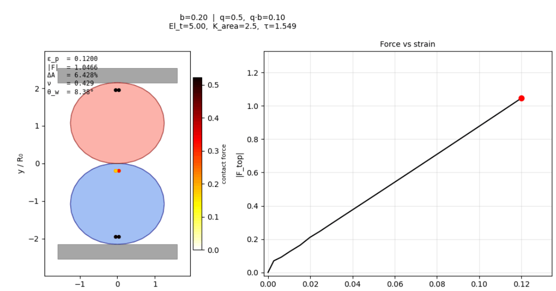


FIG. 3. **Movie S3** (`movies/S3_q0p5_compressible.gif`): Single squeeze at $q = 0.5$ ($q \cdot b = 0.10$, $\nu_{\mathrm{meas}} \approx 0.42$, $\Delta A \approx 4.3\%$). Moderate area loss; the disk begins to bulge laterally. The transition from predominantly area-compressible to mixed behavior is visible in the rounding of the equatorial cross-section. Still shown: maximum compression.

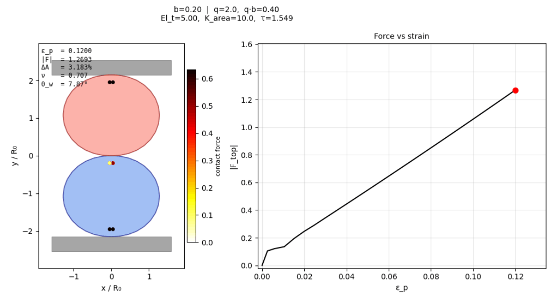


FIG. 4. **Movie S4** (`movies/S4_q2p0_intermediate.gif`): Single squeeze at $q = 2.0$ ($q \cdot b = 0.40$, $\nu_{\rm meas} \approx 0.69$, $\Delta A \approx 2.1\%$). The disk conserves area moderately well; significant lateral bulging is visible. This regime is intermediate between area-compressible and shape-deformable. Still shown: maximum compression.

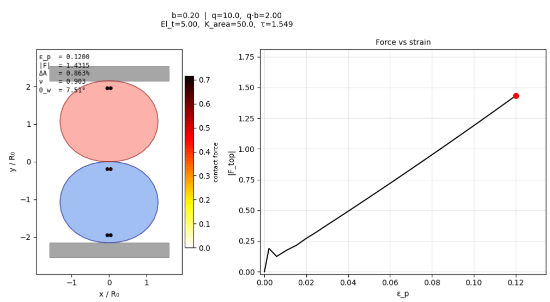


FIG. 5. **Movie S5** (`movies/S5_q10p0_shape_deformable.gif`): Single squeeze at $q = 10$ ($q \cdot b = 2.0$, $\nu_{\rm meas} \approx 0.88$, $\Delta A \approx 0.57\%$). The area is well conserved; the deformation is almost purely shape change. The disk develops a pronounced "peanut" profile under strong compression, with wide flat contact zones and equatorial bulge. Still shown: maximum compression.

MOVIE S2 — AREA-COMPRESSIBLE DISK ($q = 0.1$, $\nu_{\rm meas} \approx 0.21$)

MOVIE S3 — COMPRESSIBLE DISK ($q = 0.5$, $\nu_{\rm meas} \approx 0.42$)

MOVIE S4 — INTERMEDIATE DISK ($q = 2.0$, $\nu_{\rm meas} \approx 0.69$)

MOVIE S5 — SHAPE-DEFORMABLE DISK ($q = 10$, $\nu_{\rm meas} \approx 0.88$)

MOVIE S6 — NEARLY INCOMPRESSIBLE DISK ($q = 50$, $\nu_{\rm meas} \approx 0.94$)

MOVIES S7A–S7C — THREE-BODY COLLISIONS: MOMENTUM CONSERVATION

Movies S7a–S7c show three-body disk collision events using `step_rb()`. Two identical EPD disks approach a stationary target from opposite angles; per-frame $\Delta p_x/p_0$

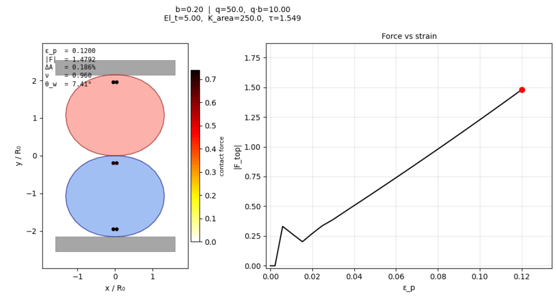


FIG. 6. **Movie S6** (`movies/S6_q50p0_nearly_incompressible.gif`): Single squeeze at $q = 50$ ($q \cdot b = 10.0$, $\nu_{\rm meas} \approx 0.94$, $\Delta A \approx 0.12\%$). Near the incompressible limit; the disk conserves area to within 0.1% and deforms entirely by shape change. The lateral bulge is maximal; the contact zone spans a large fraction of the perimeter. This is the "rubber-like" or "jello-like" end of the squishiness axis. Still shown: maximum compression.

and $\Delta p_y/p_0$ are overlaid as text. Three configurations are shown: elastic ($\alpha_d = 0$, $q = 1$), damped ($\alpha_d = 2$, $q = 1$), and rubber-like ($\alpha_d = 0$, $q = 10$). Total linear momentum is conserved to $|\Delta p|/p_0 < 2 \times 10^{-15}$ and angular momentum to $|\Delta L|/L_0 < 1.4 \times 10^{-14}$ in all cases. See `src/validation/rb_extension_benchmarks.py`.

MOVIES S8A–S8D — ELLIPSEPARTICLE: SHAPE GENERALIZATION

Movies S8a–S8d show free-flight collisions of `EllipseParticle` objects (semi-axes $a = 1.5$, $b = 0.8$, $N = 32$ equal-arc-length nodes). Four configurations are covered: E1 (equal ellipses, head-on, no spin), E2 (equal ellipses, oblique impact with initial spin $\omega_0 = \pm 0.8$), E3 (different-size oblique collision with spin), and E4 (same moment of inertia, different mass/density, with spin). Angular momentum is conserved to $|\Delta L|/L_0 \leq 8 \times 10^{-13}$ in all four cases. See `src/validation/ellipse_collision_benchmark.py`.

MOVIE S9 — EMULSION CAPILLARY-WAVE RINGDOWN

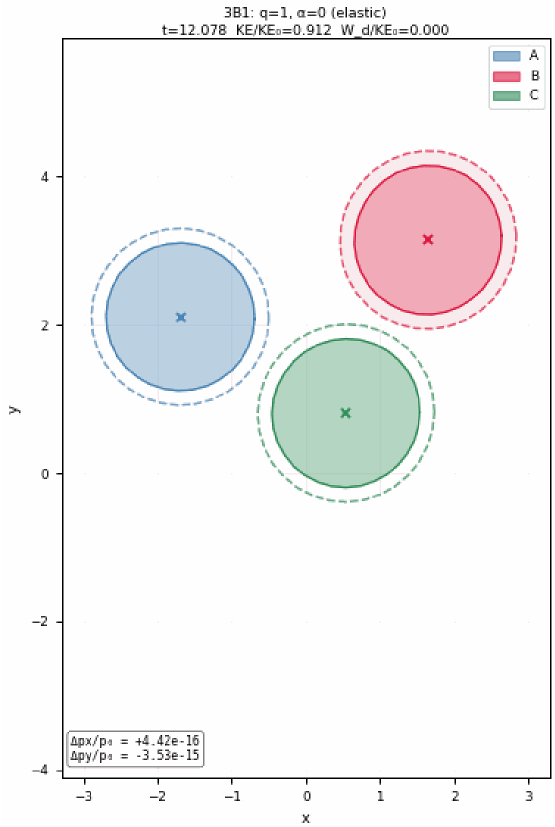
Movie S9 shows a single emulsion droplet ($N = 120$, $K_{\rm area} = 50$, $\kappa = 0.02$, $\alpha = 5.0$) released from a 10% elliptical deformation and allowed to relax. The $n = 2$ capillary-wave mode rings down over ≈ 3 oscillation periods; the exponential amplitude decay gives the measured damping rate $\alpha_{\rm meas} \approx 2.50\,\tau_0^{-1}$, consistent with the near-critical damping target $\alpha_{\rm crit} = 2\omega_2 \approx 5.46\,\tau_0^{-1}$. See `src/validation/emulsion_paper_figures.py`, Benchmark C.

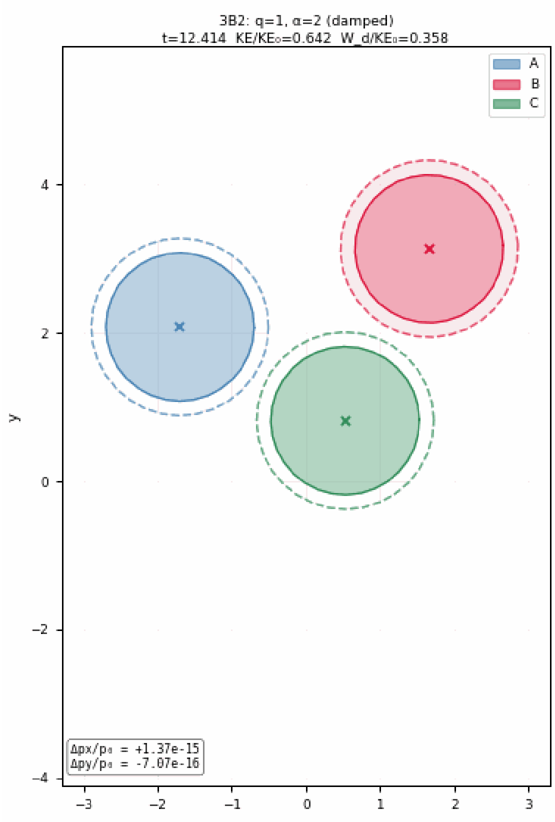
MOVIE S10 — EMULSION T1 TOPOLOGICAL REARRANGEMENT

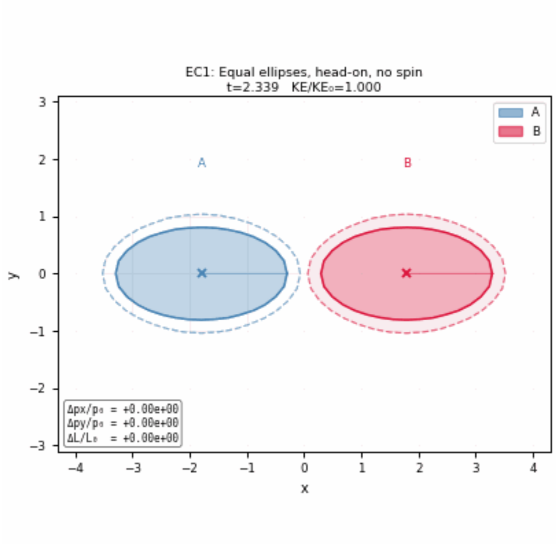
Movie S10 shows the T1-event benchmark: a center droplet squeezed vertically between two pinned neighbor droplets and two horizontal walls ($d = 1.5\,R_0$ gap, $N = 120$, $K_{\text{area}} = 50$). The center droplet elongates vertically, passes through the gap, and snaps horizontally to rest between the two neighbors — the elementary topological rearrangement of an emulsion. Area change during passage is $\Delta A < 2\%$; lateral symmetry is maintained to within $< 1\%$ of R_0 throughout. See `src/validation/emulsion_paper_figures.py`, Benchmark D.

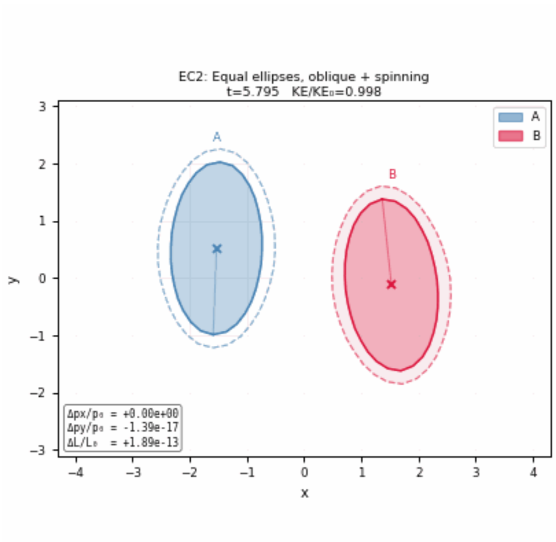
MOVIE S11 — EMULSION DROPLET FALLING UNDER GRAVITY ($\mathrm{Bo} = 0.10$)

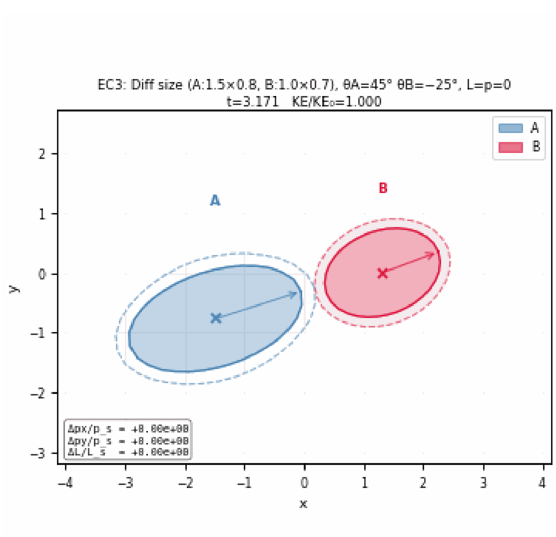
Movie S11 shows a single droplet falling freely under gravity at $\mathrm{Bo} = 0.10$ ($N = 120$, $K_{\text{area}} = 50$). The droplet accelerates from rest, reaches terminal-like behavior, and develops a visible oblate deformation: the leading (bottom) edge flattens while the trailing (top) edge rounds, giving a sag ratio $(R_{\text{top}} - R_{\text{bot}})/R_0 \approx 0.065$. Trajectory error relative to ballistic free-fall is $< 3 \times 10^{-4}$ in units of R_0. See `src/validation/emulsion_paper_figures.py`, Benchmark E.

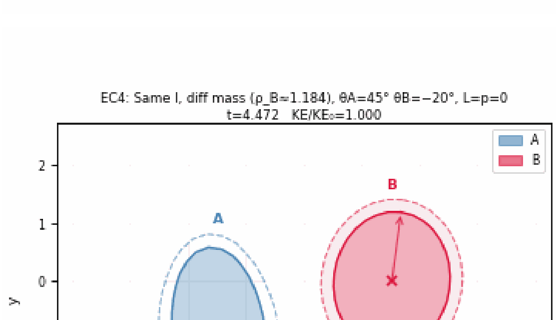

3B1: q=1, α=0 (elastic)
t=12.078 KE/KE$_0$=0.912 W_d/KE$_0$=0.000
A
B
C
Δpx/p$_0$ = +4.42e-16
Δpy/p$_0$ = -3.53e-15
x
y


3B2: q=1, α=2 (damped)
t=12.414 KE/KE$_0$=0.642 W_d/KE$_0$=0.358
A
B
C
Δpx/p$_0$ = +1.37e-15
Δpy/p$_0$ = -7.07e-16
x
y


EC1: Equal ellipses, head-on, no spin
t=2.339 KE/KE_0=1.000
A
B
$\Delta px/p_0$ = +0.00e+00
$\Delta py/p_0$ = +0.00e+00
$\Delta L/L_0$ = +0.00e+00
x
y


EC2: Equal ellipses, oblique + spinning
t=5.795 KE/KE_0=0.998
A
B
$\Delta px/p_0$ = +0.00e+00
$\Delta py/p_0$ = -1.39e-17
$\Delta L/L_0$ = +1.89e-13
x
y


EC3: Diff size (A:1.5×0.8, B:1.0×0.7), θA=45° θB=−25°, L=p=0
t=3.171 KE/KE_0=1.000
A
B
Δpx/p_s = +0.00e+00
Δpy/p_s = +0.00e+00
ΔL/L_s = +0.00e+00
x
y


EC4: Same I, diff mass (ρ_B≈1.184), θA=45° θB=−20°, L=p=0
t=4.472 KE/KE_0=1.000
A
B
y